



SAFETY DATA SHEET

TEF-CARE CONCENTRATE

MSDS No.:

AA00280-0921960000

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 테프 케어
TEF-CARE CONCENTRATE

Other means of identification : Not applicable.

Recommended use : Degreaser

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 4.8 %

Company : ECOLAB KOREA LTD.
8F, Daemyung Tower
135 Beobwon-ro, Songpa-gu, Seoul Korea 05836
82-1588-7595

Emergency telephone number : 82-1588-7595

Issuing date : 10.12.2021

2. HAZARDS IDENTIFICATION

GHS Classification

Product AS SOLD

Skin corrosion/irritation : Category 1

Serious eye damage/eye irritation : Category 1

Specific target organ toxicity - single exposure : Category 3 (Respiratory system)

Product AT USE DILUTION

Not applicable.

GHS Label element

Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements : H314 Causes severe skin burns and eye damage.
H318 Causes serious eye damage.
H335 May cause respiratory irritation.

Precautionary Statements : **Prevention:**
P260 Do not breathe mist, spray.
P264 Wash the contact area thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/ protective clothing/ eye protection/ face

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

protection.

Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.

P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or doctor/ physician.

P312 Call a POISON CENTER or doctor/ physician if you feel unwell.

P321 Specific treatment: consult MSDS Section 4.

P363 Wash contaminated clothing before reuse.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

P405 Store locked up.

Disposal:

P501 Dispose of contents and container according to wastes control act.

Product AT USE DILUTION

Hazard pictograms : Not applicable.

Signal Word : Not applicable.

Hazard Statements : Not applicable.

Precautionary Statements : **Prevention:**
P264 Wash hands thoroughly after handling.
Response:
P314 Get medical advice/ attention if you feel unwell.
Storage:
P401 Store in accordance with local regulations.

Product AS SOLD

Other hazards : None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical Name	Common name	CAS-No.	Concentration: (%)
Inorganic silicate.	no data available	1312-76-1	1 - 10
Tetrasodium EDTA	no data available	64-02-8	1 - 10
monoethanolamine	no data available	141-43-5	1 - 10
disodium metasilicate	no data available	6834-92-0	1 - 10
potassium hydroxide	no data available	1310-58-3	1 - 4.9
d-glucopyranose, oligomeric,	no data available	68515-73-1	1 - 5
decyl octyl glycosides			
sodium hydroxide	no data available	1310-73-2	1 - 4.9
Cocamidopropyl Betaine	no data available	61789-40-0	1 - 5

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

Product AT USE DILUTION

No hazardous ingredients

All unlisted components do not contribute to the overall hazard of the product and are trade secret, making up 100% of the full composition.

4. FIRST AID MEASURES

Product AS SOLD

- | | | |
|---|---|---|
| In case of eye contact | : | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately. |
| In case of skin contact | : | Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately. |
| If swallowed | : | Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately. |
| If inhaled | : | Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur. |
| Protection of first-aiders | : | If potential for exposure exists refer to Section 8 for specific personal protective equipment. |
| Notes to physician | : | Treat symptomatically. |
| Most important symptoms and effects, both acute and delayed | : | See Section 11 for more detailed information on health effects and symptoms. |

Product AT USE DILUTION

- | | | |
|-------------------------|---|---|
| In case of eye contact | : | Rinse with plenty of water. |
| In case of skin contact | : | Rinse with plenty of water. |
| If swallowed | : | Rinse mouth. Get medical attention if symptoms occur. |
| If inhaled | : | Get medical attention if symptoms occur. |

5. FIREFIGHTING MEASURES

Product AS SOLD

- | | | |
|--------------------------------------|---|---|
| Suitable extinguishing media | : | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Unsuitable extinguishing media | : | None known. |
| Specific hazards during firefighting | : | Exposure to decomposition products may be a hazard to health. |
| Hazardous combustion | : | Decomposition products may include the following materials: |

TEF-CARE CONCENTRATE

Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.
---	---

Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.
---	---

Storage temperature : 5 °C to 45 °C

4 / 12

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

Advice on safe handling	: In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
Conditions for safe storage	: Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
monoethanolamine	141-43-5	TWA	3 ppm	KR OEL
		STEL	6 ppm	KR OEL
potassium hydroxide	1310-58-3	C	2 mg/m ³	KR OEL
sodium hydroxide	1310-73-2	C	2 mg/m ³	KR OEL

Biological occupational exposure limits : no data available

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

Respiratory protection : No personal respiratory protective equipment normally required.

9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Colour	: clear, brown	light brown
Odour	: not significant	not significant
pH	: 11.2 - 12.2, (1 %)	12.5
Flash point	: Not applicable., Does not sustain combustion.	
Odour Threshold	: no data available	
Melting point/freezing point	: no data available	
Initial boiling point and boiling range	: 100 °C	
Evaporation rate	: no data available	
Flammability (solid, gas)	: Not applicable.	
Upper explosion limit	: no data available	
Lower explosion limit	: no data available	
Vapour pressure	: no data available	
Relative vapour density	: no data available	
Relative density	: 1.165 - 1.215	
Water solubility	: soluble	
Solubility in other solvents	: no data available	
Partition coefficient: n-octanol/water	: no data available	
Auto-ignition temperature	: no data available	
Thermal decomposition	: no data available	
Viscosity, kinematic	: no data available	
Explosive properties	: no data available	
Oxidizing properties	: The substance or mixture is not classified as oxidizing.	
Molecular weight	: no data available	
VOC	: no data available	

10. STABILITY AND REACTIVITY

Product AS SOLD

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

Incompatible materials	: glass Acids
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides nitrogen oxides (NOx)

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.
Ingestion	: Causes digestive tract burns.
Inhalation	: May cause respiratory tract irritation. May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

Toxicity

Product AS SOLD Product

Acute oral toxicity	: Acute toxicity estimate : > 2,000 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : > 5 mg/l Test atmosphere: dust/mist
Acute dermal toxicity	: Acute toxicity estimate : > 2,000 mg/kg
Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: no data available
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: no data available
Germ cell mutagenicity	: no data available
Teratogenicity	: no data available
STOT - single exposure	: no data available
STOT - repeated exposure	: no data available
Aspiration toxicity	: no data available

12. ECOLOGICAL INFORMATION

Toxicity

Product AS SOLD

Environmental Effects	: Harmful to aquatic life.
-----------------------	----------------------------

Product AT USE DILUTION

Environmental Effects	: This product has no known ecotoxicological effects.
-----------------------	---

Product AS SOLD Product

Toxicity to fish	: no data available
Toxicity to daphnia and other aquatic invertebrates	: no data available
Toxicity to algae	: no data available

Components

Toxicity to fish	: Inorganic silicate. 96 h LC50: 210 mg/l Tetrasodium EDTA 96 h LC50 Fish: 121 mg/l disodium metasilicate 96 h LC50 Fish: 210 mg/l Cocamidopropyl Betaine
------------------	---

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

96 h LC50 Fish: 2 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Tetrasodium EDTA
48 h EC50 Daphnia magna (Water flea): 140 mg/l

monoethanolamine
48 h LC50 Daphnia magna (Water flea): 65 mg/l

sodium hydroxide
48 h EC50: 40 mg/l

Components

Toxicity to algae : d-glucopyranose, oligomeric, decyl octyl glycosides
72 h EC50: 18 mg/l

Persistence and degradability

Product AT USE DILUTION

Poorly biodegradable

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

Ozone layer hazard: not applicable

13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : Do not contaminate storm water drains, natural waterways or soil with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (KR_DG)

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

UN number : 3266
Description of the goods : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S.
(sodium metasilicate, Potassium hydroxide)
Class : 8
Packing group : II
Environmentally hazardous : No

Sea transport (IMDG/IMO)

UN number : 3266
Proper shipping name : CORROSIVE LIQUID, BASIC, INORGANIC, N.O.S.
(sodium metasilicate, Potassium hydroxide)
Class : 8
Packing group : II
Marine pollutant : No

Special precautions for user : None

15. REGULATORY INFORMATION

National regulatory information

Industrial Safety and Health Act

Harmful Substances Prohibited from Manufacturing

Not applicable.

Korea. Harmful Substances Requiring Permission

Not applicable.

Substances to be controlled (Article 442 of the Regulation on Industrial Health Standards)

Chemical Name	CAS-No.	Threshold limits:
Ethanolamine	141-43-5	$\geq 1 \%$
Potassium hydroxide	1310-58-3	$\geq 1 \%$
Sodium hydroxide	1310-73-2	$\geq 1 \%$

Controlled Substances Subject to Environment Monitoring

Chemical Name	CAS-No.	Threshold limits:
Ethanolamine	141-43-5	$\geq 1 \%$
Potassium hydroxide	1310-58-3	$\geq 1 \%$
Sodium hydroxide	1310-73-2	$\geq 1 \%$

Controlled Substances Subject to Health Examination

Not applicable.

Substances with Workplace Exposure Limits

monoethanolamine
potassium hydroxide
sodium hydroxide

Chemicals Control Act

Toxic chemicals

Not applicable.

Authorized chemicals

Not applicable.

Restricted Toxic Chemicals

Not applicable.

Banned Toxic Chemicals

Not applicable.

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

Toxic Release Inventory Chemicals

Chemical Name	CAS-No.	Group	Threshold limits:
Potassium hydroxide	1310-58-3	Group II	$\geq 1\%$
Sodium hydroxide	1310-73-2	Group II	$\geq 1\%$

Accident Precaution Chemicals

Not applicable.

Dangerous Substances Safety Management Act:

Not Applicable to Dangerous Materials

Waste Management Law

Industrial waste

The components of this product are reported in the following inventories:

United States TSCA Inventory :

All substances listed as active on the TSCA inventory

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL.

Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :

On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :

not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :

not determined

Korea. Korean Existing Chemicals Inventory (KECI) :

On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

On the inventory, or in compliance with the inventory

China Inventory of Existing Chemical Substances :

On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :

On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet
Globally Harmonized System of Classification and Labelling of Chemicals (GHS)
IARC: (International Agency for Research on Cancer)
US. National Toxicology Program (NTP) Report on Carcinogens
ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

Issuing date : 10.12.2021
Date of first issue : 10.12.2021
version : 1.0
Prepared by : Regulatory Affairs

SAFETY DATA SHEET

TEF-CARE CONCENTRATE

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.